

GEORGINA ADAMS

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DESCRIPTION

I recently completed a master's degree in Bioinformatics at the University of Birmingham, building on my biomedical science background. I am now seeking opportunities to apply this expertise in a professional role. My academic experience includes computational biology, statistical machine learning and advanced omics technologies, including data analysis from next-generation sequencing technologies, single-cell and spatial transcriptomics. I have used and implemented R and Python throughout my degree, developing a strong technical skill set in both languages alongside a solid foundation in quantitative data analysis and statistical testing methods. My group research project is currently being developed toward co-authored publication.

Technical Skills

Programming: R, Python, Bash

Bioinformatics: Multi-omics data integration (including SGCCA, WGCNA), RNA-seq analysis (single-cell and spatial transcriptomics), biological data preprocessing and quality control, differential expression analysis, pathway and enrichment analysis, cell-type annotation and marker gene curation, cross-platform data integration

Machine Learning: Supervised and unsupervised learning, feature selection, model evaluation, clustering and classification methods.

Data Visualisation: ggplot2, matplotlib, seaborn, patchwork/cowplot for multi-panel figures, publication-quality figures, exploratory data analysis

Data Handling: Large-scale biological datasets, genomic, transcriptomic and proteomic data, reproducible workflows and statistical analysis, high-performance computing.

Scientific Communication: Thesis writing, including interpretation and presentation of complex biological data, also currently in the process of contributing as a co-author on research work conducted during one of my modules.

EMPLOYMENT HISTORY

Care Assistant, Sanctuary Care.

From: 11/2019 **To:** Present.

I have worked as a bank member of staff in care homes throughout the course of my studies since 2019, where I have worked has depended on where I lived at the time based on it being termtime/summer break.

EDUCATION

MsC Bioinformatics, University of Birmingham, Grade: TBC
From: 09/2025 To: Present

Masters Projects:

BsC Biomedical Science, Grade: First Class.

Final Year Thesis Project:

Birmingham City University.
From: 09/2022 To: 06/2025.

CertHE Adult Nursing, Grade: Pass.

Birmingham City University.
From: 01/2021 To: 12/2021

A-Level Education

Worcester Sixth Form College
From: 09/2017 To: 06/2019

Subject	Grade
Biology	C
Psychology	B
Health and Social Care	A*

GCSEs Education

9 GCSEs Grades A-C including English and Maths.

WORK EXPERIENCE

Nursing Placement (Adult Nursing). Good Hope Hospital.

From: September 2021 **To:** December 2021 (12 weeks).

This was completed as part of my Adult Nursing degree, I decided to leave the degree after year 1, and went on to do my Biomedical Science degree. However, I gained valuable insight into patient-centered care within an acute hospital setting and developed my understanding of clinical practice, multidisciplinary teamwork, and healthcare delivery. Skills I gained from this included:

- Teamwork in multidisciplinary settings
- Clinical documentation and professionalism
- Understanding of clinical workflows

Projects

MSc Dissertation

Context-Dependent Reliability of Cell-Type Markers in Single-Cell and Spatial Transcriptomics
(Undertaken at Birmingham Tissue Analytics)

University of Birmingham | [May/26] - [Aug/26]

- Curated and validated a 42-marker panel for five immune/endothelial liver lineages from single-cell RNA-seq data, using combined statistical (log2FC, AUC) and visual validation criteria
- Independently assessed cross-platform marker transferability to CosMx spatial transcriptomics (1,000- and 6,000-gene panels), identifying panel coverage and detection sensitivity as primary constraints on transfer, rather than marker quality
- Built a full R/Seurat analysis pipeline (QC, clustering, differential expression, cross-platform statistical comparison) and produced publication-style figures using ggplot2, patchwork and custom visualisation tools

GROUP RESEARCH PROJECT

Transcriptomic analysis of seasonal gene expression in *Quercus robur*

UNIVERSITY OF BIRMINGHAM | [Oct/26] - [May/26]

I conducted a comprehensive RNA-seq differential expression analysis investigating the molecular basis of flowering regulation in pedunculate oak (*quercus robur*) under ambient and elevated co₂ conditions, using data generated at the Birmingham institute of forest research (BiFor) face facility. I was responsible for most of the bioinformatics pipeline. This work is currently being prepared for publication. This involved the following:

- Preprocessing of raw sequencing data using Fastqc, Trimmomatic, hisat2, Samtools, and HTseq-count differential expression analysis in r using DESeq2, including seasonal pairwise comparisons, eco₂ vs ambient within-month comparisons, and juvenile vs adult developmental comparisons
- Generation of all visualisations including volcano plots, upset plots, heatmaps, and venn diagrams compilation of priority candidate gene lists for downstream go enrichment and blast annotation.

Tools and languages: R, DESeq2, ggplot2, pheatmap, upsetr, ggvenn, hisat2, samtools, htseq, fastqc, trimmomatic, bluebear hpc, GitHub

MSc Coursework: Mathematical & Statistical Analysis using R

Applied statistical analysis and reproducible R workflow

UNIVERSITY OF BIRMINGHAM | [Oct/25] - [DEC/25]

- Built a reproducible R workflow to analyse probability distributions, statistical inference, and mathematical models, applying core techniques used in bioinformatics data analysis.
- Designed modular, reusable R functions to streamline statistical computation, improve code clarity, and support scalable, maintainable analysis pipelines.
- Conducted hypothesis testing, distribution fitting, and numerical analysis, presenting results through structured, publication-style visualisations.

BSc Dissertation

Computational Identification of Bioportides Targeting EFCAB9 for Non-Hormonal Male Contraception

- Built a bioinformatics workflow to identify the sperm-specific target protein EFCAB9, isolate its key binding motif, and design two candidate peptide therapeutics (bioportides), then validated their efficacy, safety, and structural stability entirely in silico.
- Skills gained: database mining and integration, target identification and validation, peptide/protein sequence design, interpretation of machine learning prediction outputs, protein structure prediction and visualisation, in silico safety/efficacy evaluation.

Tools used: DisGeNET, UniProt, Human Protein Atlas (HPA), Protein Data Bank (PDB), Complex Portal, MLCPP 2.0, InterPro, PepCalc, Immune Epitope Database (IEDB), AlphaFold, ChimeraX.